

Problem Set 2:

Submission: 12:00 Fri, Apr 24, 2026; Solutions: Mon, Apr 27, 2026

Problem 1 (Zentralübung) The one-dimensional classical string as a free scalar field theory

The goal of this exercise is to show how a classical field theory emerges naturally when considering a system with infinitely many particles. Specifically, we will be considering a system of point masses connected together by springs and then we will let the distance of these point masses go to zero while keeping their number density constant.

Let us start by considering a pair of two points, each with mass m . The spring that connects both points is characterized by its spring constant k and creates a harmonic potential between the points. Furthermore, let $\bar{\phi}_0, \bar{\phi}_1$ denote the x -coordinates of the two points relative to their equilibrium positions. There are no degrees of freedom in the y - or z -directions.

(1.a) Give an expression for the kinetic energy T and the potential energy V of that two-point system and state the equations of motion (no need to solve them).

Next, we want to add another $N - 2$ more points, connected with identical springs. We want to 'connect' the two ends of the string, meaning that the string is *closed*. The oscillators are constrained to vibrate along the circumference of a ring, and the periodic constraint is fully encoded in $\bar{\phi}_i = \bar{\phi}_{i+N}$ for all $i = 0, 1, \dots, N - 1$. Here, $\bar{\phi}_i$ denotes the linear position of point mass i as before.

(1.b) Generalize your result from part (a) to a periodic system of N particles, and combine both to find the Lagrangian $L = T - V$ and Hamiltonian $H = T + V$ of the system.

Now, our goal is to take the continuum limit by letting the average distance ℓ between adjacent point masses go to zero, such that the total length $L = N\ell$, the mass per unit length $\mu = m/\ell$, and the string tension $T = k\ell$ are constant. While taking that limit, we venture the transition from discrete coordinates to a continuous field $\phi(x, t)$, where

$$\bar{\phi}_i(t) = \bar{\phi}(x_i = i\ell, t) \rightarrow \bar{\phi}(x, t). \quad (1)$$

(1.c) Use this replacement in conjunction with

$$\ell \sum_{i=0}^{N-1} \xrightarrow{\ell \rightarrow 0} \int_0^L dx, \quad (2)$$

$$(\bar{\phi}_{i+1} - \bar{\phi}_i)/\ell \xrightarrow{\ell \rightarrow 0} \partial_x \bar{\phi}(x, t), \quad (3)$$

to derive the Lagrangian and Hamiltonian *densities* $\mathcal{L}$ and $\mathcal{H}$ defined by

$$H(t) = \int_0^L dx \mathcal{H}(x, t), \quad (4)$$

$$L(t) = \int_0^L dx \mathcal{L}(x, t). \quad (5)$$

Within special relativity, time and space are treated on equal footing. To manifest this in our results for the Lagrangian and Hamiltonian densities, we would like to rescale the field by the string tension:

$$\bar{\phi} \rightarrow \phi = \bar{\phi} \sqrt{T}. \quad (6)$$

At the same time, it is convenient to rescale the time coordinate such that it has the same dimension as the spatial coordinate. To this end, we introduce the wave velocity $v^2 \equiv T/\mu$ and rescale

$$t \rightarrow t' = vt. \quad (7)$$

(1.d) Convince yourself that both the Lagrangian and Hamiltonian densities do not explicitly depend on the parameters μ, T after rescaling both the time and the field.

(1.e) Derive the equations of motion for the field ϕ by varying $\phi \rightarrow \phi + \delta\phi$ in the action

$$S[\phi, \partial_x \phi, \partial_t \phi] = \int dt L(t). \quad (8)$$

We have thus shown how a free scalar field theory emerges from a very simple classical system after taking an appropriate continuum limit. When this field theory is supplied with additional interactions such as a quartic interaction ϕ^4 , it has real physical applications and can, for example, be understood as the continuum limit of the Ising model.

Solution:

(1.a)

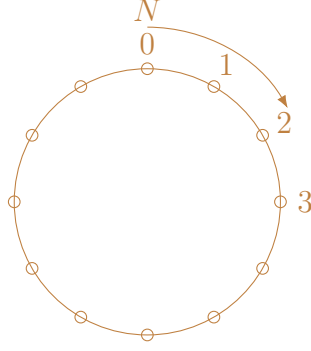
$$T = \frac{1}{2}m \left[\left(\frac{d\bar{\phi}_0}{dt} \right)^2 + \left(\frac{d\bar{\phi}_1}{dt} \right)^2 \right] \quad (9)$$

$$V = \frac{1}{2}k (\bar{\phi}_1 - \bar{\phi}_0)^2 \quad (10)$$

Equation of motion

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\phi}_i} - \frac{\partial \mathcal{L}}{\partial \phi_i} = 0, \quad \text{with} \quad \mathcal{L} = T - V \quad (11)$$

$$\begin{cases} m \ddot{\bar{\phi}}_0 + k(\bar{\phi}_0 - \bar{\phi}_1) = 0, \\ m \ddot{\bar{\phi}}_1 + k(\bar{\phi}_1 - \bar{\phi}_0) = 0. \end{cases} \quad (12)$$



(1.b) Periodic boundary conditions:

$$\phi_0 = \phi_N \quad (\phi_i = \phi_{i \in N} \quad \forall i) \quad (13)$$

$$T = \frac{1}{2}m \sum_{i=0}^{N-1} \left(\frac{d\bar{\phi}_i}{dt} \right)^2 = \frac{1}{2}m \sum_{i=0}^{N-1} \dot{\bar{\phi}}_i^2 \quad (14)$$

and

$$V = \frac{1}{2}k \sum_{i=0}^{N-1} (\bar{\phi}_{i+1} - \bar{\phi}_i)^2 \quad (15)$$

$$\mathcal{L} = T - V = \frac{1}{2}m \sum_{i=0}^{N-1} \dot{\bar{\phi}}_i^2 - \frac{1}{2}k \sum_{i=0}^{N-1} (\bar{\phi}_{i+1} - \bar{\phi}_i)^2 \quad (16)$$

$$H = T + V = \frac{1}{2}m \sum_{i=0}^{N-1} \dot{\bar{\phi}}_i^2 + \frac{1}{2}k \sum_{i=0}^{N-1} (\bar{\phi}_{i+1} - \bar{\phi}_i)^2 \quad (17)$$

(1.c)

$$\bar{\phi}_i(t) = \bar{\phi}(x_i = i\ell, t) \rightarrow \bar{\phi}(x, t) \quad (18)$$

with

$$\ell \rightarrow 0 \quad \text{such that} \quad L = N\ell, \quad \mu = \frac{m}{\ell}, \quad T = k\ell \quad \text{remain constant.}$$

Then

$$\ell \sum_{i=0}^{N-1} \xrightarrow{N \rightarrow \infty} \int_0^L dx \quad (19)$$

and

$$\frac{\bar{\phi}_{i+1} - \bar{\phi}_i}{\ell} \longrightarrow \partial_x \bar{\phi}(x_i, t). \quad (20)$$

For the kinetic term:

$$\frac{1}{2}m \sum_{i=0}^{N-1} \dot{\bar{\phi}}_i^2 = \frac{1}{2} \underbrace{\frac{m}{\ell}}_{=\mu} \ell \sum_{i=0}^{N-1} (\partial_t \bar{\phi}(x_i, t))^2 \longrightarrow \frac{1}{2}\mu \int_0^L dx (\partial_t \bar{\phi}(x, t))^2. \quad (21)$$

For the potential term:

$$\frac{1}{2}k \sum_{i=0}^{N-1} (\bar{\phi}_{i+1} - \bar{\phi}_i)^2 = \frac{1}{2} \underbrace{k\ell}_{=T} \sum_{i=0}^{N-1} \left(\frac{\bar{\phi}_{i+1} - \bar{\phi}_i}{\ell} \right)^2 \longrightarrow \frac{1}{2}T \int_0^L dx (\partial_x \bar{\phi}(x, t))^2. \quad (22)$$

Thus

$$L = \int_0^L dx \mathcal{L}(x, t) \quad \text{with} \quad \mathcal{L}(x, t) = \frac{1}{2}\mu (\partial_t \bar{\phi})^2 - \frac{1}{2}T (\partial_x \bar{\phi})^2. \quad (23)$$

(1.d) We rescale the field

$$\bar{\phi} \rightarrow \phi = \sqrt{T} \bar{\phi} \quad (24)$$

and the time coordinate

$$t \rightarrow vt = t', \quad v = \sqrt{\frac{T}{\mu}}, \quad (25)$$

such that

$$[x] = [t] = \text{“length”}. \quad (26)$$

Then

$$\partial_x \bar{\phi} = \frac{1}{\sqrt{T}} \partial_x \phi, \quad \partial_t \bar{\phi} = \frac{v}{\sqrt{T}} \partial_{t'} \phi = \frac{1}{\sqrt{\mu}} \partial_{t'} \phi. \quad (27)$$

Thus the Lagrangian density becomes

$$\mathcal{L}(t', x, \phi, \partial_{t'} \phi, \partial_x \phi) = \frac{1}{2} [(\partial_{t'} \phi)^2 - (\partial_x \phi)^2]. \quad (28)$$

(1.e)

$$S[\phi, \partial_x \phi, \partial_{t'} \phi] = \int_{t'_0}^{t'_1} dt' L(t') = \int dt' dx \mathcal{L}(x, t'). \quad (29)$$

Principle of least action:

$$\delta S = 0. \quad (30)$$

From this follows

$$\delta \int dt' dx \mathcal{L}(x, t') = 0 \quad (31)$$

and therefore

$$\int dt' dx \left[\frac{\partial \mathcal{L}}{\partial \phi} \delta \phi + \frac{\partial \mathcal{L}}{\partial (\partial_{t'} \phi)} \delta (\partial_{t'} \phi) + \frac{\partial \mathcal{L}}{\partial (\partial_x \phi)} \delta (\partial_x \phi) \right] = 0. \quad (32)$$

If we assume that $\delta \phi(t'_0) = \delta \phi(t'_1) = 0$, then all boundary terms vanish (and x has periodic boundary conditions), hence

$$\int dt' dx \left[\frac{\partial \mathcal{L}}{\partial \phi} - \partial_{t'} \frac{\partial \mathcal{L}}{\partial (\partial_{t'} \phi)} - \partial_x \frac{\partial \mathcal{L}}{\partial (\partial_x \phi)} \right] \delta \phi = 0. \quad (33)$$

Therefore, the Euler–Lagrange equation holds:

$$\partial_{t'} \frac{\partial \mathcal{L}}{\partial(\partial_{t'} \phi)} + \partial_x \frac{\partial \mathcal{L}}{\partial(\partial_x \phi)} = 0, \quad (34)$$

and thus

$$\boxed{\partial_{t'}^2 \phi - \partial_x^2 \phi = 0.} \quad (35)$$

Comparison with the discrete case:

$$m \partial_t^2 \phi_i + k(\phi_i - \phi_{i+1}) + k(\phi_i - \phi_{i-1}) = 0. \quad (36)$$

In the continuum this becomes

$$m \partial_t^2 \phi_i \longrightarrow m \partial_t^2 \phi(x_i, t) = \mu \ell \partial_t^2 \phi(x_i, t), \quad (37)$$

and

$$\begin{aligned} k(\phi_{i+1} - 2\phi_i + \phi_{i-1}) &= k \left(\frac{\phi_{i+1} - \phi_i}{\ell} - \frac{\phi_i - \phi_{i-1}}{\ell} \right) \ell \\ &\longrightarrow k \ell^2 \partial_x^2 \phi(x_i, t) = \ell T \partial_x^2 \phi(x_i, t). \end{aligned}$$

Thus

$$\mu \ell \partial_t^2 \phi(x_i, t) - \ell T \partial_x^2 \phi(x_i, t) = 0, \quad (38)$$

or equivalently

$$\mu \partial_t^2 \phi(x, t) - T \partial_x^2 \phi(x, t) = 0. \quad (39)$$

These are the same equations of motion.

Problem 2 Electrostatic Theorems

Suppose a charge distribution $\rho_1(\mathbf{r})$ creates a potential $\varphi_1(\mathbf{r})$, and some other charge distribution $\rho_2(\mathbf{r})$ produces a potential $\varphi_2(\mathbf{r})$. Let us emphasize that both charge distributions are *not* present at the same time, we are talking about two different problems. Then the following relation holds:

$$\int d^3r \rho_2(\mathbf{r}) \varphi_1(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int d^3r \int d^3r' \frac{\rho_2(\mathbf{r}) \rho_1(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} = \int d^3r' \varphi_2(\mathbf{r}') \rho_1(\mathbf{r}') \quad (40)$$

(2.a) Use the relation (40) to prove that the average of $\varphi(\mathbf{r})$ over the surface of an empty sphere S of radius R is equal to the potential at the center of the sphere:

$$\varphi(0) = \frac{1}{4\pi R^2} \int_{\partial S} dA \varphi(\mathbf{r}) \quad (41)$$

Hint: Introduce an empty sphere with charge spread uniformly over its surface as a “reciprocal system”, for which the potential inside and outside the surface is known to be

$$\varphi_2(\mathbf{r}) = \frac{q}{4\pi\epsilon_0} \begin{cases} \frac{1}{R}, & r \leq R, \\ \frac{1}{r}, & r \geq R. \end{cases} \quad (42)$$

(2.b) Next, use Green’s second identity,

$$\int_V d^3r (f \nabla^2 g - g \nabla^2 f) = \int_{\partial V} d\mathbf{A} \cdot (f \nabla g - g \nabla f), \quad (43)$$

to provide an alternative proof of the mean-value identity (41).

Our identity (41) can be used to prove another, much more general statement:

The scalar potential $\varphi(\mathbf{r})$ in a finite, charge-free region of space V takes its extremum values on the boundary ∂V of V .

(2.c) Give a proof of the above statement using the mean-value identity (41).

Hint: We recommend an indirect proof (show that there cannot be local extrema in V°). Remember that the interior $V^\circ = V \setminus \partial V$ is characterized by the fact that for all $x \in V^\circ$ there exists a finite radius ϵ such that the spherical ball $B_\epsilon(x)$ of radius ϵ centered around x is still contained in the interior.

Solution:

(2.a) As suggested in the problem, we choose a uniformly charged sphere as a reference system:

$$\rho_2(\mathbf{r}') = \frac{q}{4\pi R^2} \delta(r' - R). \quad (44)$$

Let ρ_1 and φ_1 be the charge density and potential of the sphere S . Then

$$\int d^3r' \rho_2(\mathbf{r}') \varphi_1(\mathbf{r}') = \frac{q}{4\pi R^2} \int_S dS \varphi_1(\mathbf{r}') = q \langle \varphi \rangle_S. \quad (45)$$

On the other hand, $\rho(\mathbf{r}'') = \rho_1(\mathbf{r}'')$ is non-zero only outside the sphere of radius R :

$$\int d^3\mathbf{r}'' \rho_1(\mathbf{r}'') \varphi_2(\mathbf{r}'') = \frac{q}{4\pi\epsilon_0} \int_{r''>R} d^3\mathbf{r}' \frac{\rho(\mathbf{r}')}{r'} = q \varphi(0). \quad (46)$$

$$\implies \langle \varphi \rangle_S = \varphi(0). \quad (47)$$

(2.b) Let the sphere S have radius R and its center at the origin. Since its interior is charge free, we have

$$\nabla^2 \varphi(\mathbf{r}) = 0 \quad \text{for } r < R. \quad (48)$$

Define

$$f = \varphi(\mathbf{r}), \quad g = \frac{1}{r} - \frac{1}{R}. \quad (49)$$

Furthermore, introduce

$$A_\epsilon := \{\mathbf{r} : \epsilon < r < R\}, \quad (50)$$

which is a sphere of radius R , with the inside cut out until radius ϵ . Inside of A_ϵ , we have

$$\nabla^2 f = \nabla^2 g = 0. \quad (51)$$

Using Green's identity:

$$\int_{\partial A_\epsilon} (\varphi \nabla g) \cdot d\mathbf{A} = \int_{\partial A_\epsilon} (g \nabla \varphi) \cdot d\mathbf{A}. \quad (52)$$

Let $B_a(0)$ denote a ball of radius a centered around the origin. Note that

$$g|_{\partial B_R(0)} = 0 \quad \text{and} \quad \nabla g|_{\partial B_R(0)} = -\hat{\mathbf{e}}_r \frac{1}{R^2}. \quad (53)$$

Furthermore,

$$\nabla g|_{\partial B_\epsilon(0)} = -\hat{\mathbf{e}}_r \frac{1}{\epsilon^2}, \quad g|_{\partial B_\epsilon(0)} = \frac{1}{\epsilon} - \frac{1}{R}. \quad (54)$$

Then:

$$\begin{aligned} \int_{\partial A_\epsilon} (\varphi \nabla g) \cdot d\mathbf{A} &= \int_{\partial B_\epsilon(0)} \varphi(\mathbf{r}') \left(-\hat{\mathbf{e}}_r \frac{1}{\epsilon^2} \right) \cdot d\mathbf{A} \\ &\quad + \int_{\partial B_R(0)} \varphi(\mathbf{r}') \left(-\hat{\mathbf{e}}_r \frac{1}{R^2} \right) \cdot \hat{\mathbf{e}}_r dA \\ &\xrightarrow{\epsilon \rightarrow 0} 4\pi\varphi(0) - 4\pi\langle \varphi \rangle_S. \end{aligned} \quad (55)$$

$$\begin{aligned} \int_{\partial A_\epsilon} (g \nabla \varphi) \cdot d\mathbf{A} &= \int_{\partial B_\epsilon(0)} \left(\frac{1}{\epsilon} - \frac{1}{R} \right) \nabla \varphi \cdot d\mathbf{A} + 0 \\ &= \mathcal{O}(\epsilon). \end{aligned} \quad (56)$$

$$\implies \varphi(0) = \langle \varphi \rangle_S. \quad (57)$$

To get the signs right, it is important to remember that

$$d\mathbf{A} = -\hat{\mathbf{e}}_r dA \quad (58)$$

at the inner boundary of A_ϵ .

(2.c) Wlog, let $x \in V^\circ$ be a minimum. Choose $\varepsilon > 0$ such that $B_\varepsilon(x) \subset V^\circ$.

Then

$$\varphi(x) = \langle \varphi \rangle_{B_\varepsilon(x)} \quad (59)$$

by using the mean-value identity.

$$\implies \exists y \in B_\varepsilon(x) : \varphi(y) \leq \varphi(x). \quad (60)$$

$$\implies \forall \tilde{\varepsilon} < \varepsilon \exists y \in V^\circ \text{ with } |y - x| < \tilde{\varepsilon} : \varphi(y) \leq \varphi(x), \quad (61)$$

contradicting the assumption that x is a local minimum.

Problem 3 Electrostatics of a Charged Disk

Consider a two-dimensional disk of radius R with a uniform charge per unit area $\sigma > 0$. For concreteness, choose the origin to coincide with the center of the disk, and orient it such that the disk lies in the $x - y$ -plane.

(3.a) Calculate the potential at any point *on the z -axis*.

(3.b) Calculate the potential at any point on the rim of the disk

Hint: Now it may be useful to use a point on the rim as the origin.

(3.c) Sketch the electric field pattern everywhere in the plane of the disk.

(3.d) Calculate the electrostatic total energy

$$U_E = \frac{1}{2} \int d^3r \rho(\mathbf{r}) \varphi(\mathbf{r}). \quad (62)$$

of the disk.

Solution:

(3.a)

$$\varphi(z) := \frac{1}{4\pi\varepsilon_0} \int d^3\mathbf{r}' \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} \Big|_{\mathbf{r}=(0,0,z)}^\top \quad (63)$$

$$\text{with } \rho(\mathbf{r}') = \sigma \delta(z') \Theta(R - \varrho'). \quad (64)$$

$$\Rightarrow \varphi(z) = \frac{\sigma}{4\pi\varepsilon_0} 2\pi \int_0^R d\varrho \varrho \frac{1}{\sqrt{\varrho^2 + z^2}} \quad (65)$$

$$= \frac{\sigma}{2\varepsilon_0} \left(\sqrt{z^2 + R^2} - |z| \right). \quad (66)$$

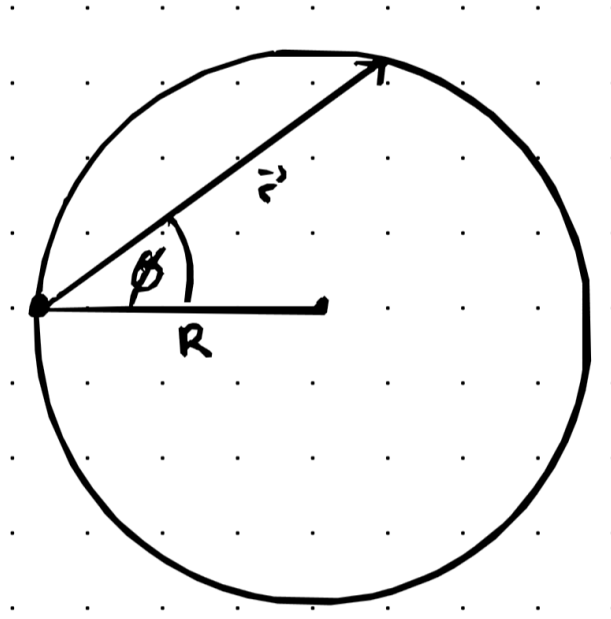


Abbildung 1: Geometry of the integration in part (3.b)

(3.b) Now consider a point on the rim of the disk and use it as the origin.

$$0 \leq |\mathbf{r}| \leq 2R \cos \phi \quad \text{for} \quad \phi \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right). \quad (67)$$

The potential at that point is therefore

$$\varphi_{\text{rim}} = \frac{\sigma}{4\pi\epsilon_0} \int_{-\pi/2}^{+\pi/2} d\phi \int_0^{2R \cos \phi} dr \, r \frac{1}{r} \quad (68)$$

$$= \frac{\sigma}{4\pi\epsilon_0} \int_{-\pi/2}^{+\pi/2} d\phi \, 2R \cos \phi = \frac{\sigma R}{\pi\epsilon_0}. \quad (69)$$

(c) The electric field is radially symmetric and points outwards by symmetry. For $r \gg R$, it behaves like a point charge. At $r = 0$, it is zero by symmetry.

$$\mathbf{E}(\mathbf{0}) = \mathbf{0}. \quad (70)$$

$$\|\mathbf{E}(\mathbf{r})\| \sim \frac{1}{r^2} \quad \text{for } r \gg R. \quad (71)$$

Qualitatively, $\|\mathbf{E}(\mathbf{r})\|$ rises from 0, reaches a maximum near $r = R$, and then decays like $1/r^2$.

(3.d) Let

$$\rho(\mathbf{r}) = \sigma \Theta(R - r) \delta(z),$$

and let

$$D \equiv \{\mathbf{r} \in \mathbb{R}^3 \mid x^2 + y^2 \leq R^2, z = 0\}$$

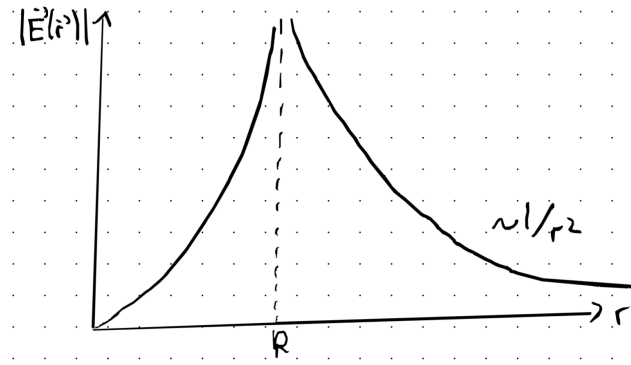


Abbildung 2: Magnitude of the electric field in the plane of the disk

be the disk.

We set

$$\frac{1}{4\pi\epsilon_0} = 1.$$

Then

$$\begin{aligned} U_E &= \frac{1}{2} \int d^3\mathbf{r} \int d^3\mathbf{r}' \frac{\rho(\mathbf{r})\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} \\ &= \frac{\sigma^2}{2} \int_D dx dy \int_D dx' dy' \frac{1}{|\mathbf{r} - \mathbf{r}'|} \Big|_{\mathbf{r}=(x,y,0), \mathbf{r}'=(x',y',0)}. \end{aligned}$$

Using the change of variables

$$\mathbf{u} \equiv \mathbf{r} - \mathbf{r}', \quad \mathbf{r}' \equiv \mathbf{v},$$

we have

$$\mathbf{v} \in D \quad \text{and} \quad \mathbf{u} + \mathbf{v} \in D,$$

so that

$$\mathbf{v} \in D \cap (D - \mathbf{u}).$$

Moreover,

$$\mathbf{u} \in B_{2R}(0) = D - D = D + D,$$

with $B_{2R}(0)$ denoting a two-dimensional ball of radius $2R$ in the xy -plane, centered around the origin. Hence,

$$U_E = \frac{\sigma^2}{2} \int_{B_{2R}(0)} d^2\mathbf{u} \frac{1}{u} \int_{D \cap (D - \mathbf{u})} d^2\mathbf{v}.$$

The inner integral can be evaluated geometrically. Using Fig. 3, one finds

$$\theta = \arccos\left(\frac{u}{2R}\right).$$

Half of the area of the intersection $|D \cap (D - \mathbf{u})|$ is the difference of a circular arc of angle 2θ and the blue triangle in Fig. 3.

$$|D \cap (D - \mathbf{u})| = 2R^2 \arccos\left(\frac{u}{2R}\right) - u\sqrt{R^2 - \frac{u^2}{4}}.$$

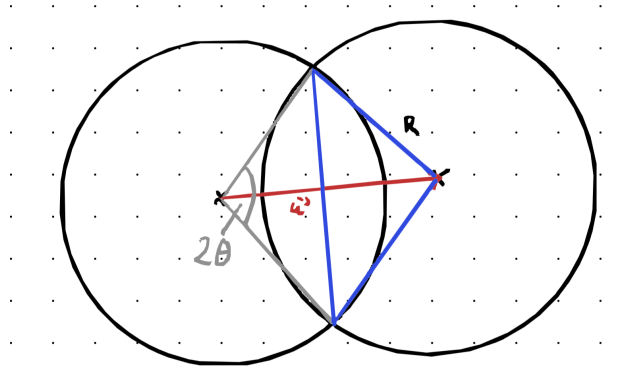


Abbildung 3: Sketch for the derivation of $|D \cap (D - \mathbf{u})|$

Thus,

$$\begin{aligned}
 U_E &= \frac{\sigma^2}{2} \cdot 2\pi \int_0^{2R} du \left(2R^2 \arccos\left(\frac{u}{2R}\right) - u\sqrt{R^2 - \frac{u^2}{4}} \right) \\
 &= \frac{\sigma^2}{2} \cdot 2\pi \cdot \frac{8R^3}{3} \\
 &= \frac{8\pi\sigma^2 R^3}{3}.
 \end{aligned}$$